GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code: DI05023041

Subject Name: Advanced Plastic Materials

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-03

Prerequisite:	Student should have knowledge regarding properties of various commodity plastic materials and engineering plastic materials
Rationale:	In today's plastic-age, to satisfy the need of end-user, many advanced polymers have been developed. Diploma plastic engineer has to deal with the production as a responsible technician and first line supervisor in the industries. Thus he should be acquainted with various properties of: specialty polymers, alloys & blends of polymers to deal with advanced applications of such materials. To comply with environmental issues particularly: plastic waste, biodegradable polymer works as a solution. This course has been design to develop a capacity of the learner to select polymer for a specific application.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Select appropriate material for a specific application.	U,A
02	Compare various properties among specific polymeric groups.	R,U
03	Select the Blend components.	U,A
04	Compare properties of different biopolymers.	R,U
05	Select a proper plastic material for appropriate applications of the sector.	U,A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Specialty Polymers	Properties and Applications of: 1.1 Polyimides (PI) 1.2 Polyetherimide (PEI) 1.3 Poly(amide-imide) (PAI) 1.4 Polysulphone (PSO) 1.5 Polyethersulphone (PES) 1.6 Polyphenylenesulphide (PPS) 1.7 Polyetheretherketone (PEEK)	08	18
Unit – II Specific Polymeric groups	Properties & Applications of : 2.1 High Temperature Polymers 2.2 Fire Resisting Polymers 2.3 Liquid Crystal Polymers (LCP) 2.4 Electro-active Polymers 2.4.1 Inherently Conductive Polymers 2.4.2 Photoconductive Polymers 2.4.3 Polymers in Fiber Optics 2.5 Thermoplastic Elastomers (TPEs) 2.6 Hydrogels and Smart Polymers 2.7 Wood Polymer Composites (WPC)	09	20
Unit – III Polymer Alloys & Blends	3.1 Introduction & definitions of terms related to Alloys & Blends. 3.2 Importance of Polymer alloys & blends 3.3 Selection of blend components. 3.4 Steps in polymer blend design. 3.5 Blend Preparation techniques 3.5.1 Melt blending 3.5.2 Solution blending 3.5.3 Latex or dispersion mixing 3.6 Properties and applications of alloys and blends: 3.6.1 PP/EPDM 3.6.2 PC/ABS 3.6.3 PC/PBT (Xenoy) 3.6.4 PPO/Nylon 3.6.5 PPO/PS (Noryl) 3.6.6 ABS/PVC	10	22

Unit – IV Biodegradable Polymers	4.1 Introduction & definition of related terms. 4.2 Classification of biodegradable polymers 4.3 Properties and applications of biodegradable polymers: 4.3.1 Natural polymers (starch , cellulose) 4.3.2 Microbial polyesters (PHA,PHB) 4.3.3 Polylactic acid (PLA) 4.3.4 Polycaprolactone (PCL) 4.3.5 Water Soluble Polymers 4.3.6 Degradable Polymers 4.3.6.1Packaging Application 4.3.6.2Medical and Related Applications 4.4 Factors affecting biodegradability 4.5 Processing of biodegradable polymers (Injection & Extrusion)	10	22
Unit – V Trends in Polymer Applications	Polymer's applications in the sector of: 5.1 Packaging 5.2 Building and Construction 5.3 Automotive Applications 5.4 Aerospace Applications 5.5 Agriculture 5.6 Medical and Biomedical Applications 5.7 Sports	08	18
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
25	**35**	**40**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S.No.	Title of Book	Author	Publication with place, year and ISBN
1	Hand Book Biodegradable Polymers	Catia Bastioli	Rapra Technology Limited, 2005, 9781859573891
2	Green Plastics	Stevens	Princeton University Press, 2001, 9780691049670

3	Biodegradable polymers for industrial applications	Ray Smith	Woodhead Publishing Limited, 2005, 9780849334665
4	Plastics Materials	Brydson	Butterworth-Heinemann, 1999, 9780750641326
5	Polymer Alloys & Blends	R.P.Singh	Asian Books, 2002, 9788186299296
6	Engineering Thermoplastics	James Margolis	CRC Press, 1985, 9780849326462
7	Plastics Technology Handbook	Manas Chanda & Roy	CRC Press, 2017, 9781498786218
8	Principles of Polymer Science	Bahadur & Sastry	Alpha Science International, 2007, 9788173196553
9	Polymers Blends & Alloys	L.A.Uttracki	Hanser Publications, 1990, 9780195207965
10	Engineering Polymers	Dyson	Chapman & Hall, 1990, 9780412020810

(b) Open-source software and website:

1. http://cdn.intechopen.com/pdfs-wm/34065.pdf
2. http://ed.iitm.ac.in/~shankar_sj/Courses/ED5312/Materials_for_Automobiles17.pdf
3. http://www.icmpp.ro/sustainableplastics/files/Biodegradable_plastics_and_polymers.pdf
4. https://www.scribd.com/presentation/629640130/Liquid-Crystal-Ppt-1
5. https://www.slideshare.net/slideshow/biodegradable-polymers-25995228/25995228?utm_source=clipboard_share_button&utm_campaign=slideshare_make_sharing_viral_v2&utm_variation=control&utm_medium=share
6. https://www.scribd.com/document/801019640/Unit-1-Speciality-Polymers-Engineering-Chemistry

33. Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	Select a plastic material among PI, PAI & PEI for aircraft component application.	02
2	Select a specialty polymer, based on its properties, for sliding component product.	02
3	Select suitable polymer for the rechargeable battery.	04
4	Prepare a product of Wood Plastic Composite (WPC).	04
5	Select blend components to make efficient alloy & blend for automobile product.	02

6	Select blend components to make efficient alloy & blend for washing machine part.	02
7	Select PLA as a bio-degradable polymer for eco-friendly purpose.	04
8	Select bio-degradable polymer for eco-friendly application.	04
9	Select polymer for agriculture & medical applications.	04
10	Select polymer for sports applications.	02

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Students will visit nearby advanced plastic manufacturing industry.
4. Fetch new polymers used in specialty applications.
5. Identify new areas of applications of polymers.
6. Prepare list of properties and applications of specialty polymers.
7. Prepare list of properties and applications of polymer alloys and blends.
8. Explore the importance of biodegradable polymers and latest trends in it with the help of internet.

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